

A 77GHz 8-bit CMOS Phase Shifter Adopting a Nested-Vector-Based Error Correction with 0.33°/0.07dB RMS-Error for MIMO Radar Applications

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Outline

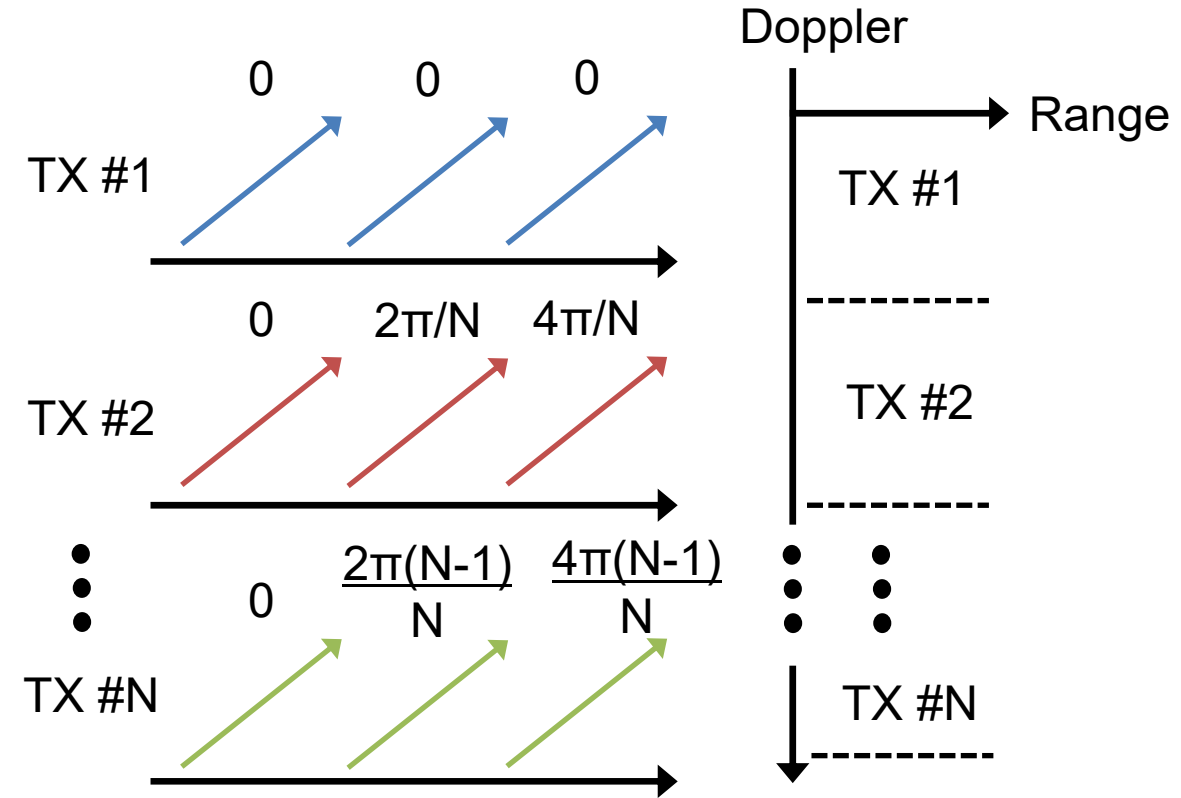
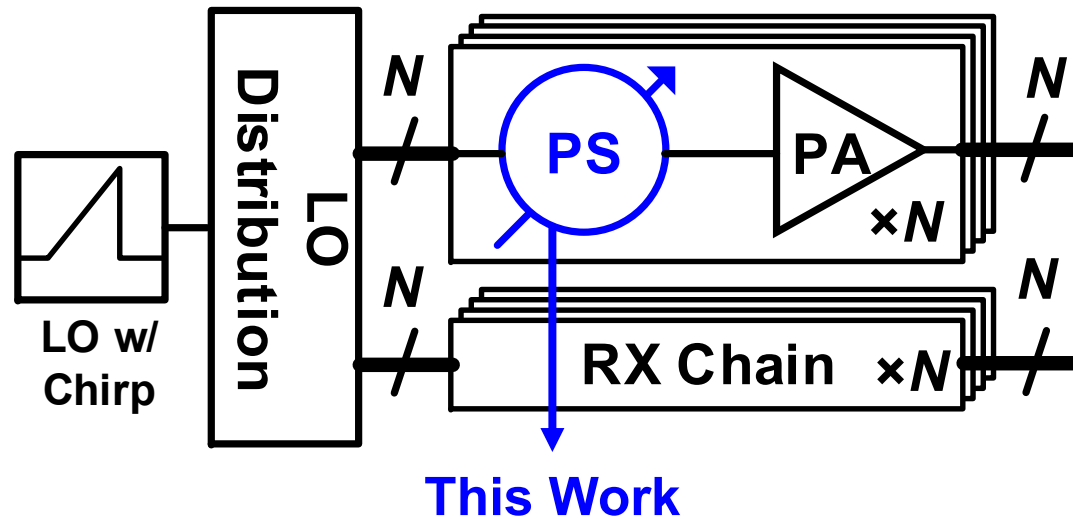
- **Motivation**
- **Proposed Architecture**
- **Circuit Implementation**
- **Measurement Results**
- **Conclusion**

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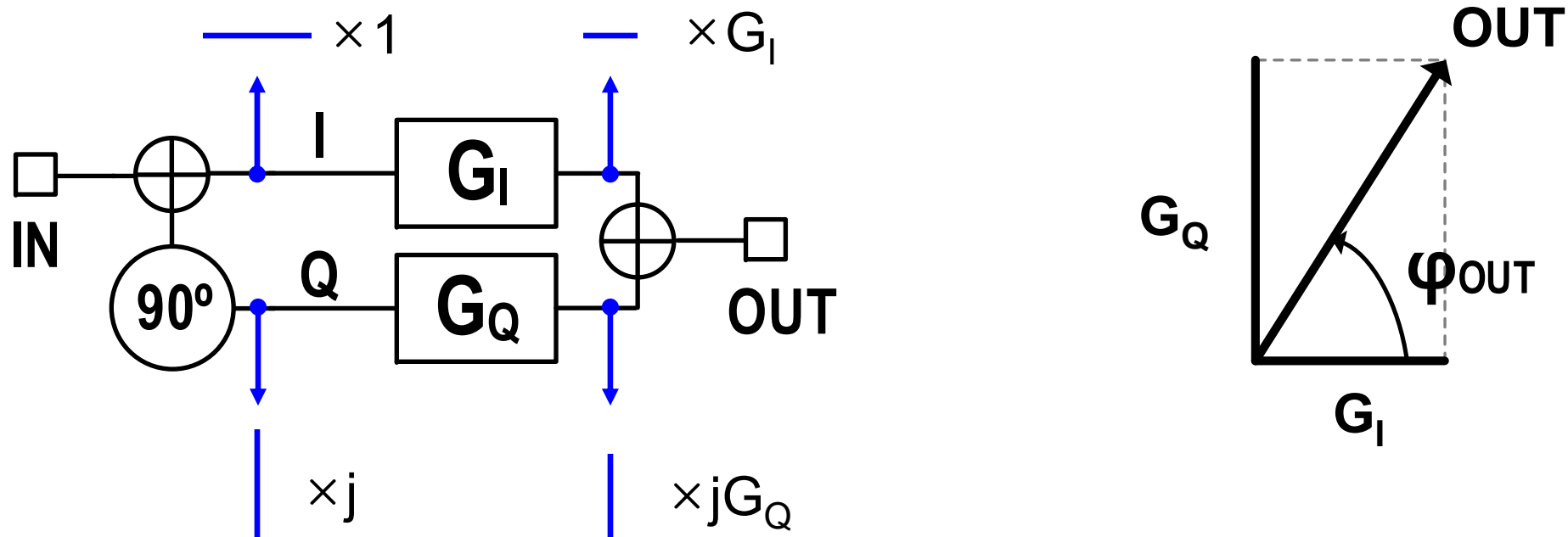
Motivation

mmW MIMO Radar TRx
(will be presented in 34.2)



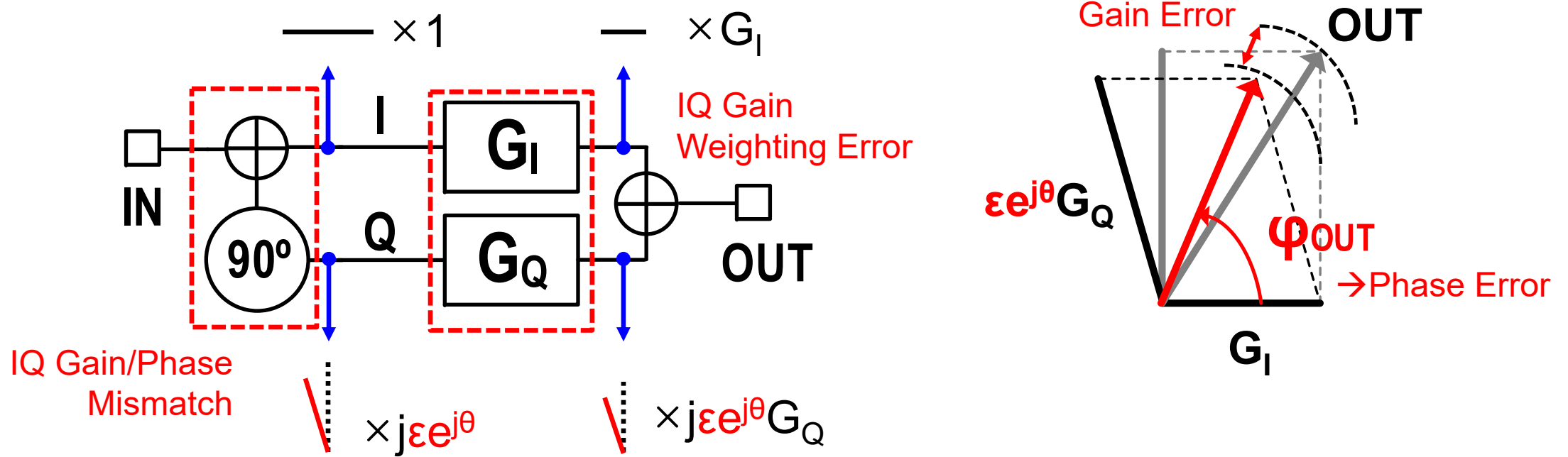
- Millimeter-wave radar RFIC is one of the key technologies for ADAS
- A High resolution PS enables to better TX orthogonality in Doppler Division Multiplexing (DDM) MIMO

Vector-Sum Phase Shifters (VSPSs)



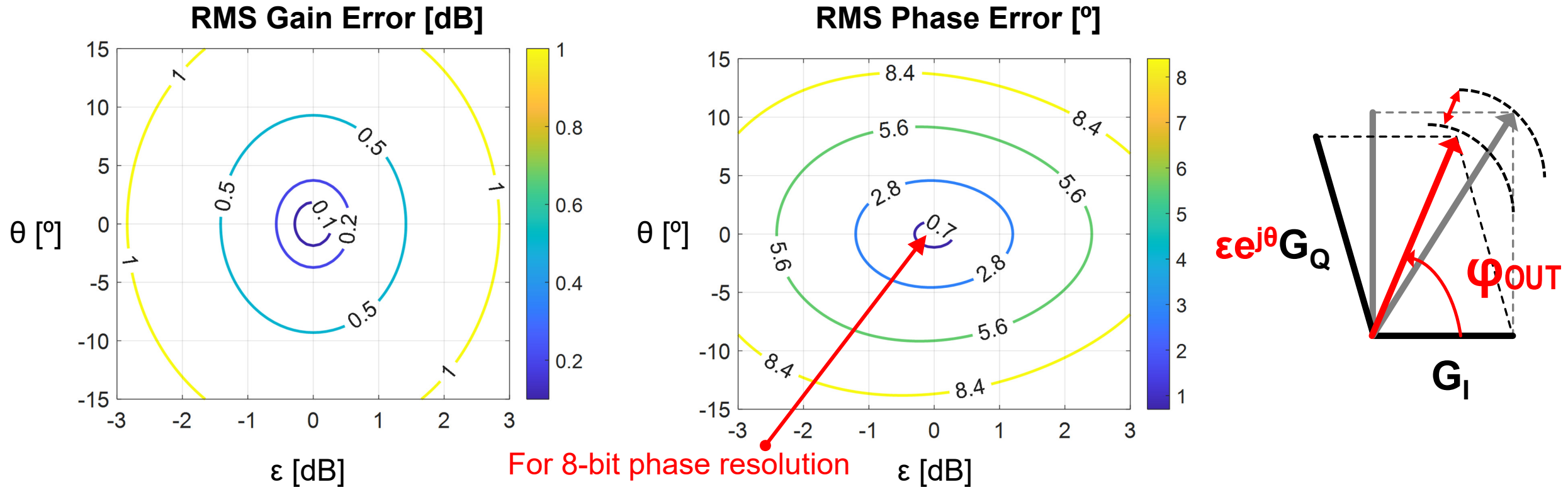
- Low loss, compact
- High phase resolution, phase code calibration possible

Vector-Sum Phase Shifters (VSPSs)



- Error sources: IQ Mismatch (IQMM), Gain weighting error
- Limited phase resolution

Vector-Sum Phase Shifters (VSPSs)



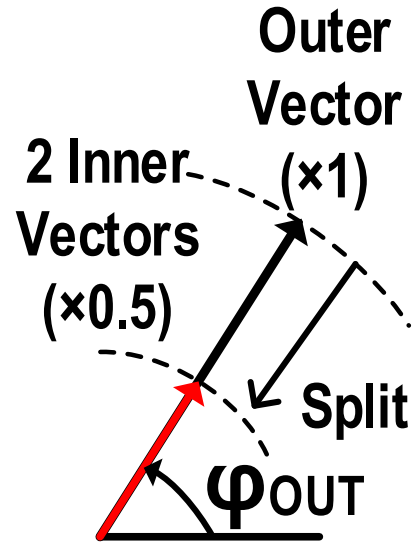
- Very challenging to achieve 8-bit resolution over wideband
- Advanced technique is needed for high resolution PS

Outline

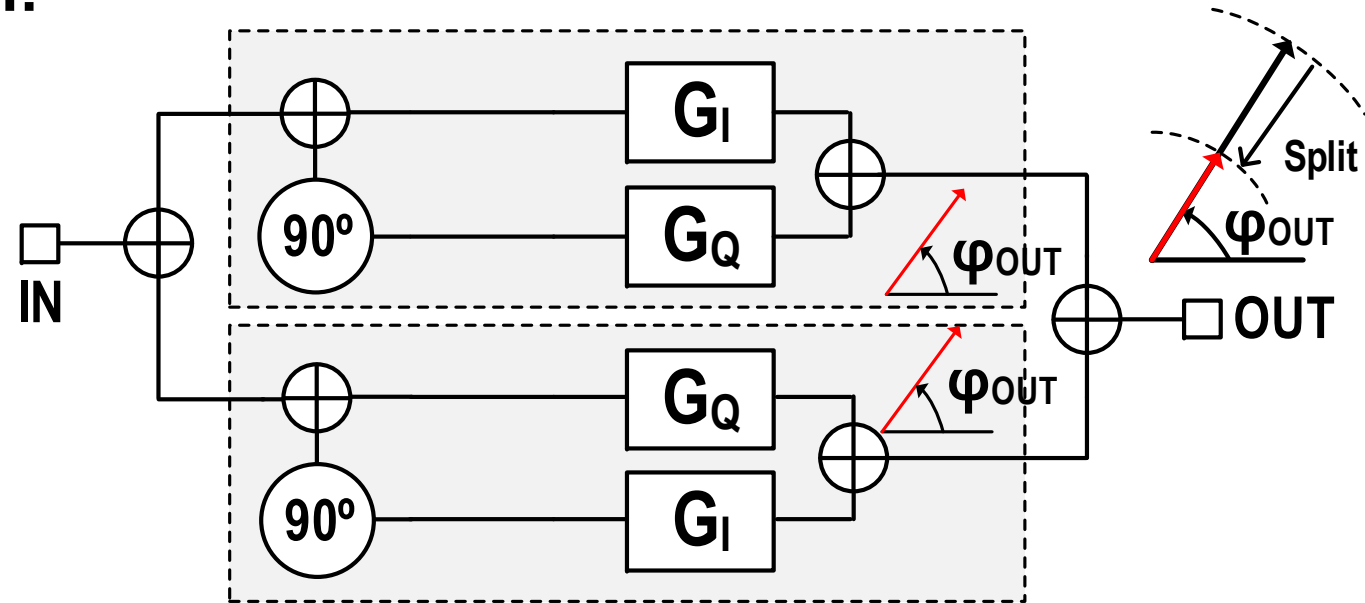
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Proposed Nested VSPS

Concept:



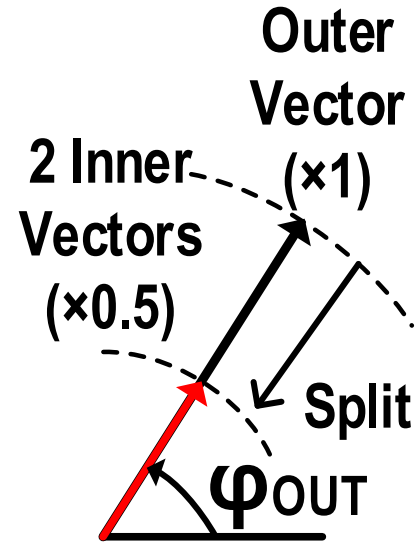
Realization:



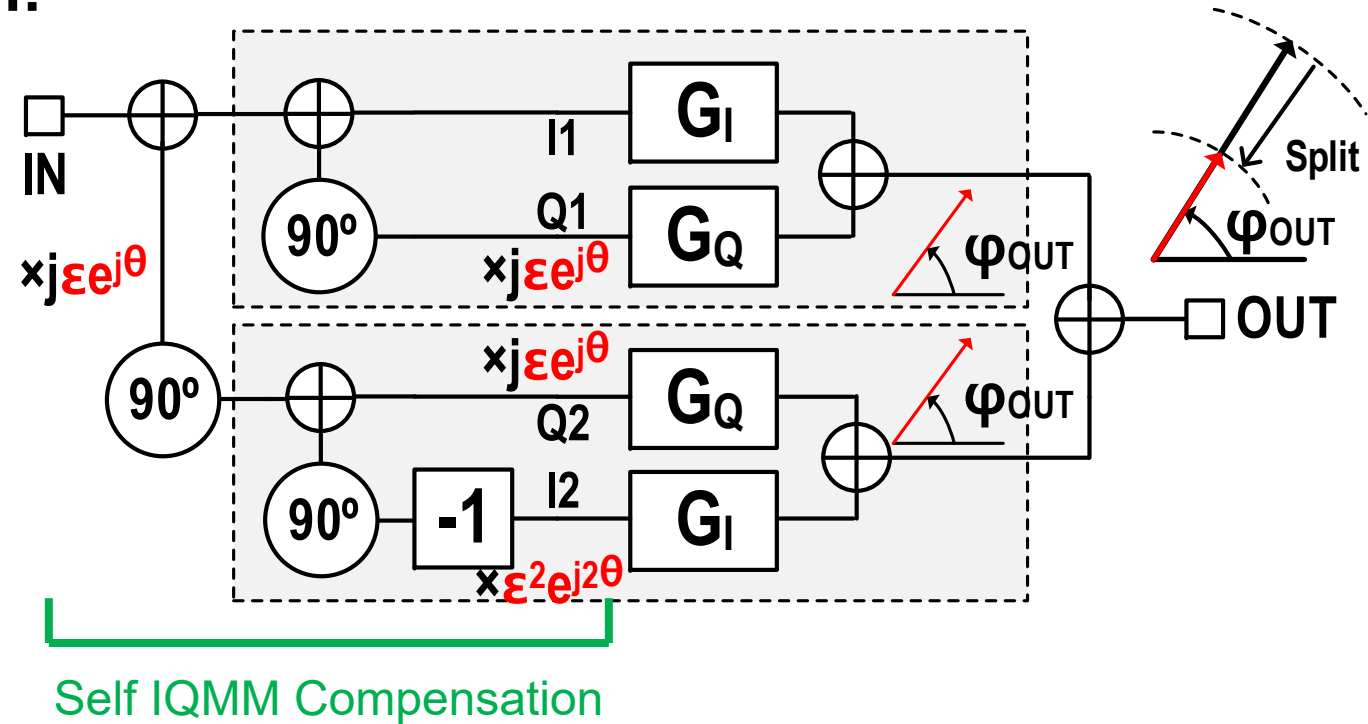
- Basic idea: decomposition of the outer vector into two inner vectors
- Two parallel VSPSs for two modulated inner vectors

Proposed Nested VSPS

Concept:



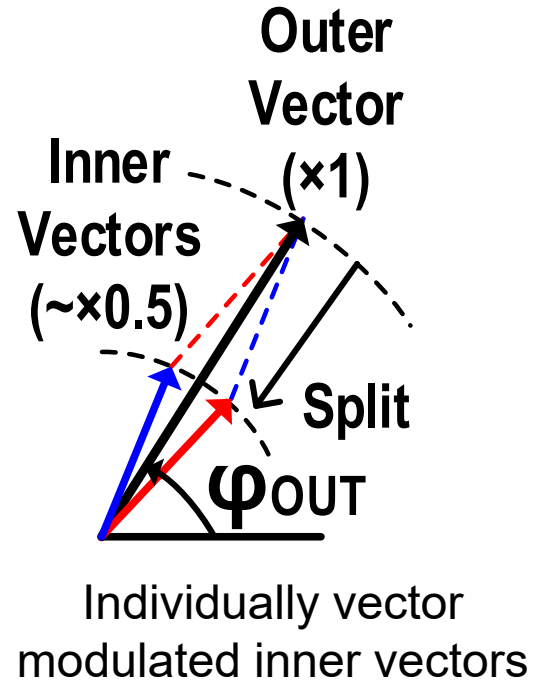
Realization:



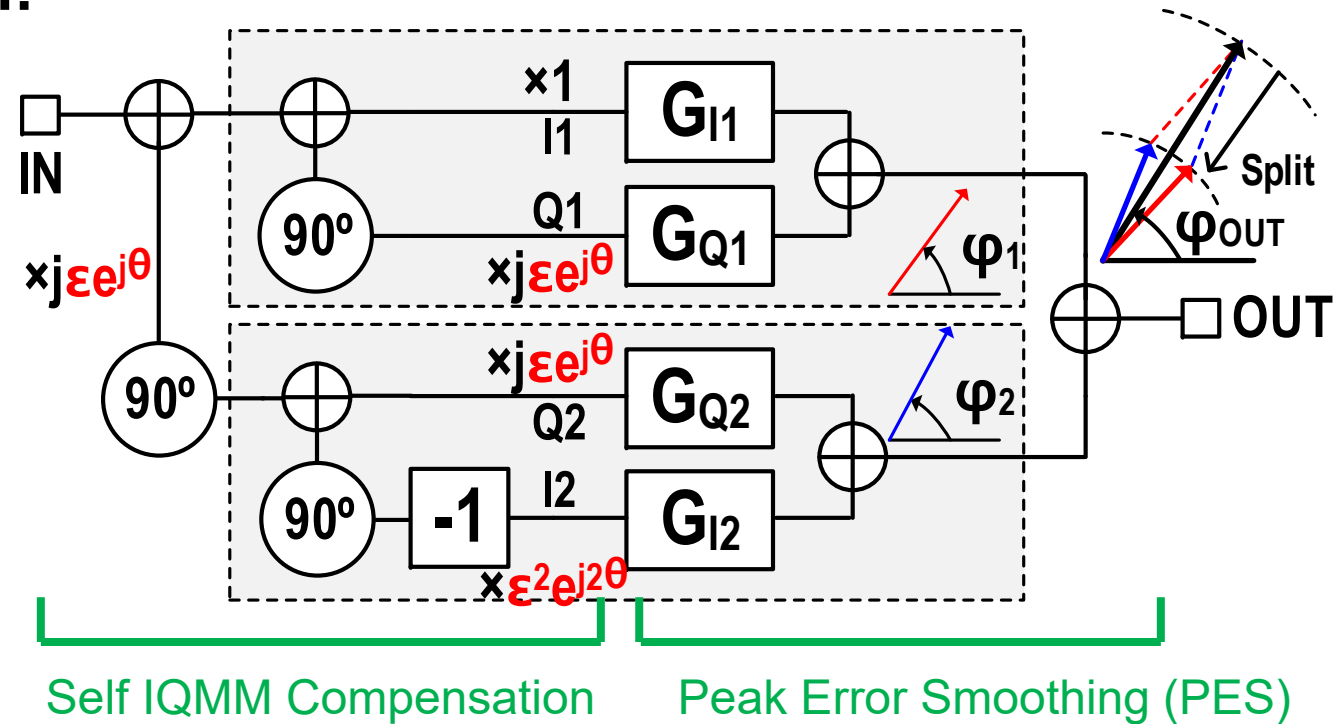
- Point 1: Leveraging multi-stage IQ generation for multiple inner vector generation
- Assume that each IQ generator is identical

Proposed Nested VSPS

Concept:



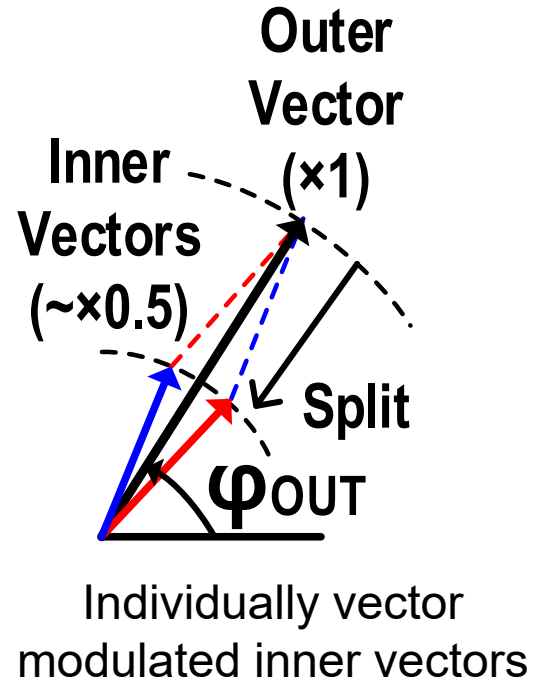
Realization:



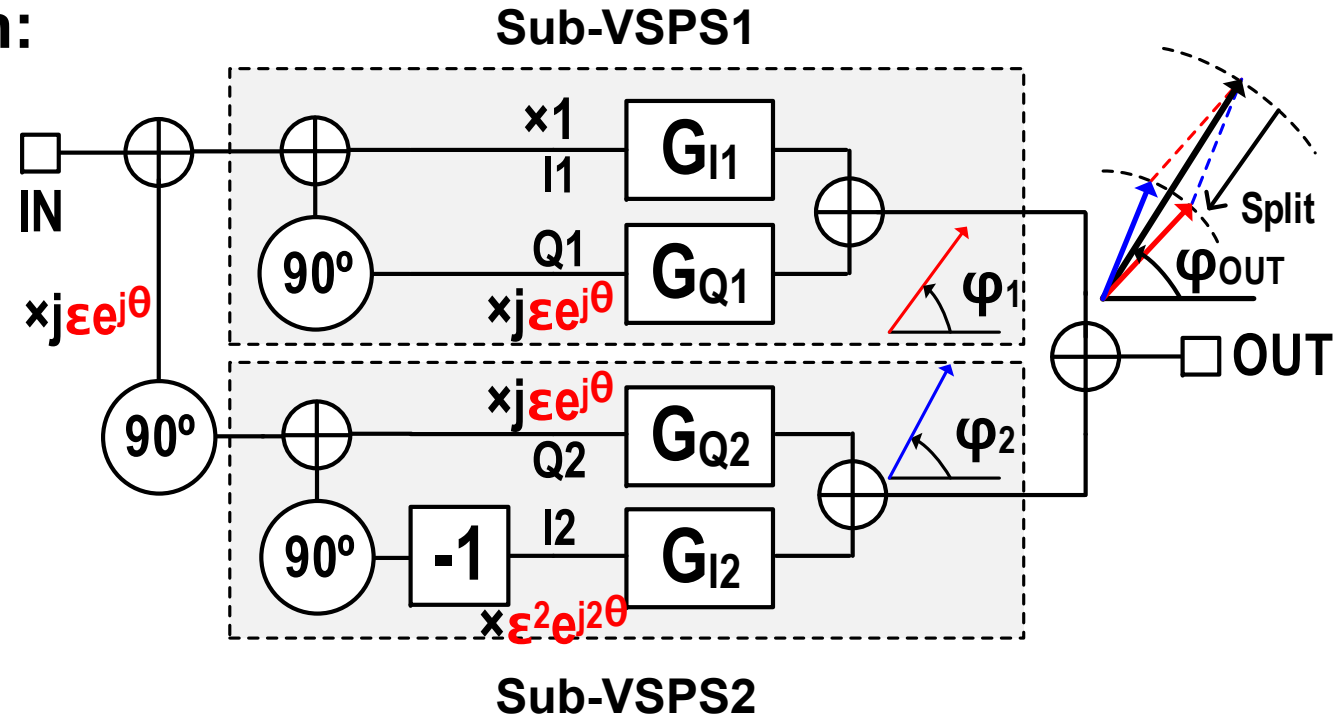
- **Point 2: Two inner vectors are individually vector modulated for vector error reduction → Peak Error Smoothing (PES)**

Proposed Nested VSPS

Concept:



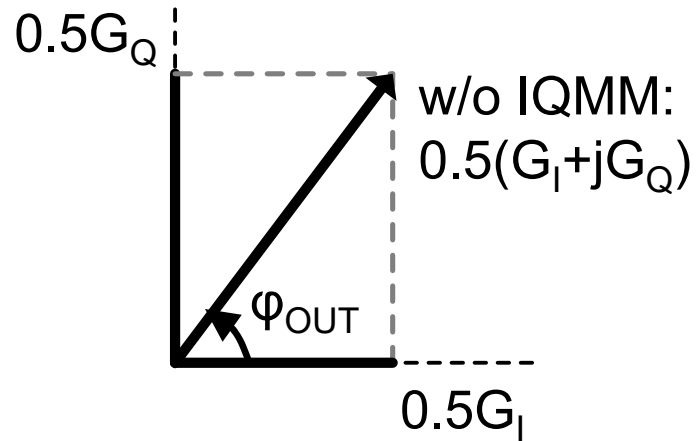
Realization:



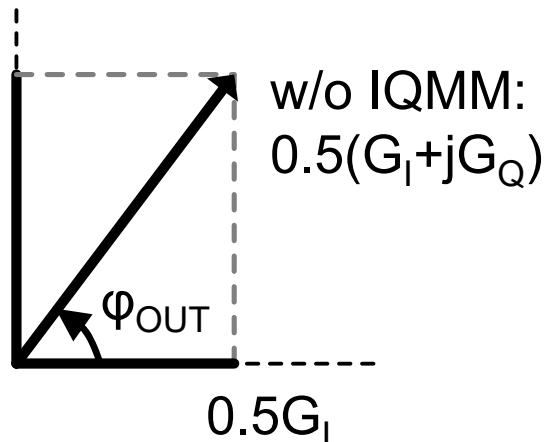
- Proposed VSPS is nested architecture of two VSPSs → Nested VSPS

Nested Vector Modulation w/o IQMM

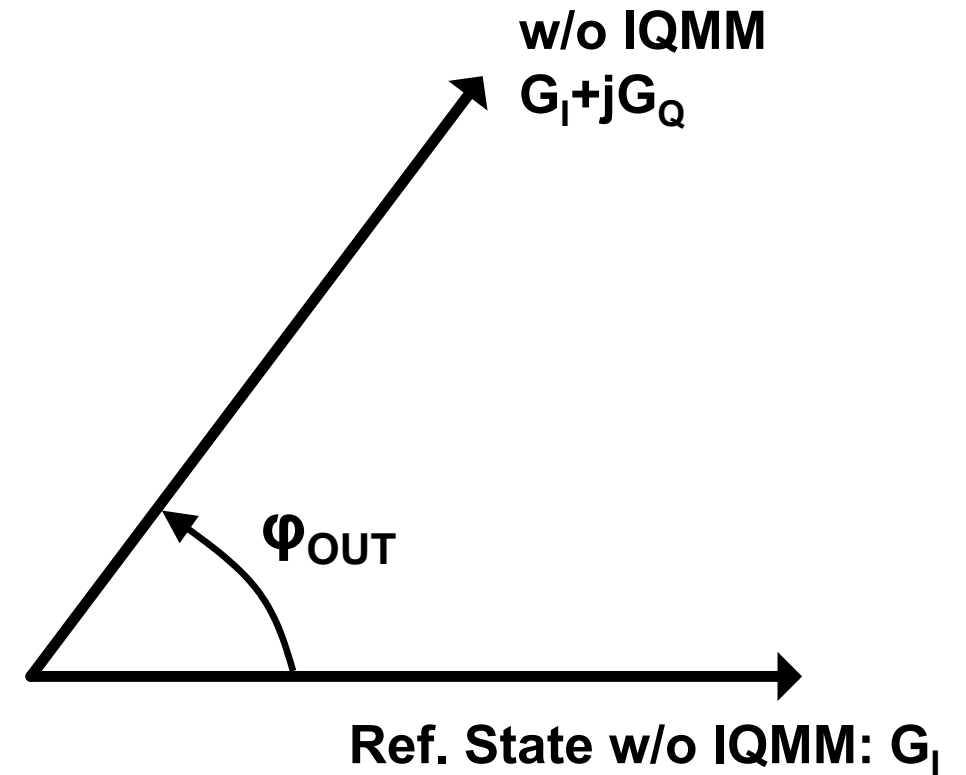
Sub-VSPS1:



Sub-VSPS2:

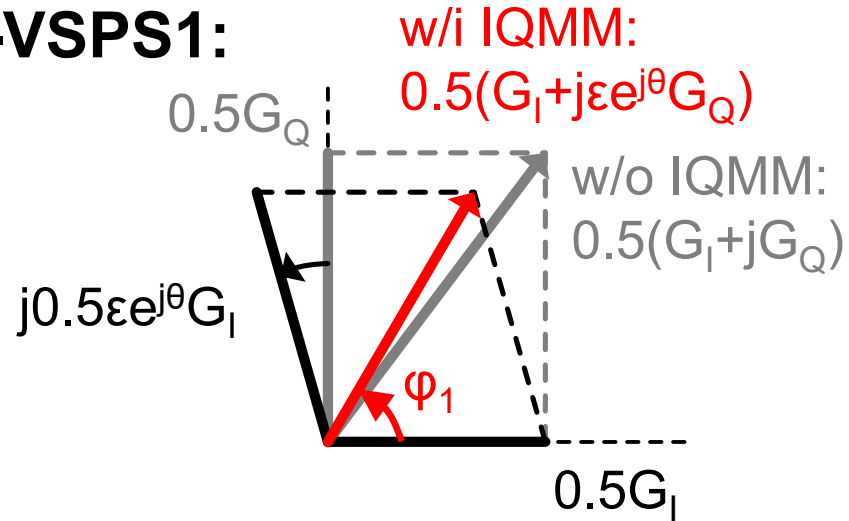


Nested VSPS Output:

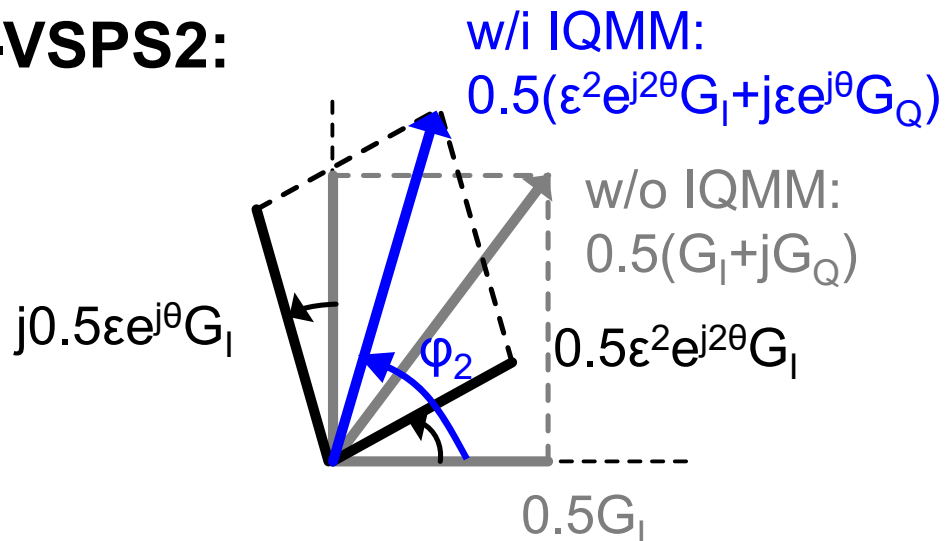


Nested Vector Modulation with IQMM

Sub-VSPS1:



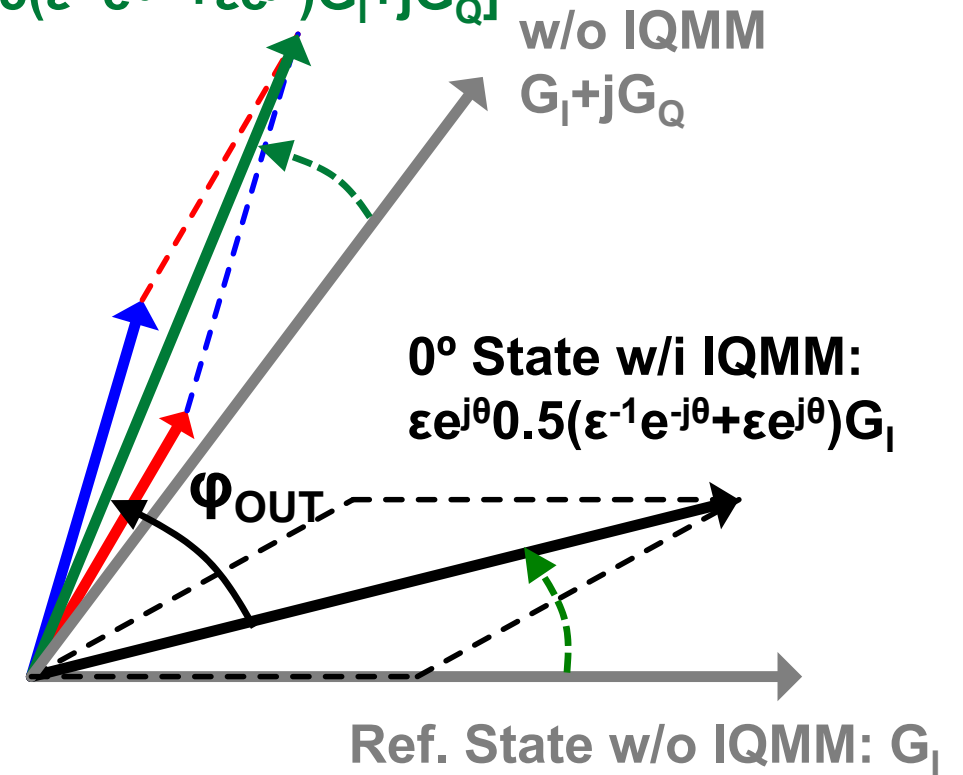
Sub-VSPS2:



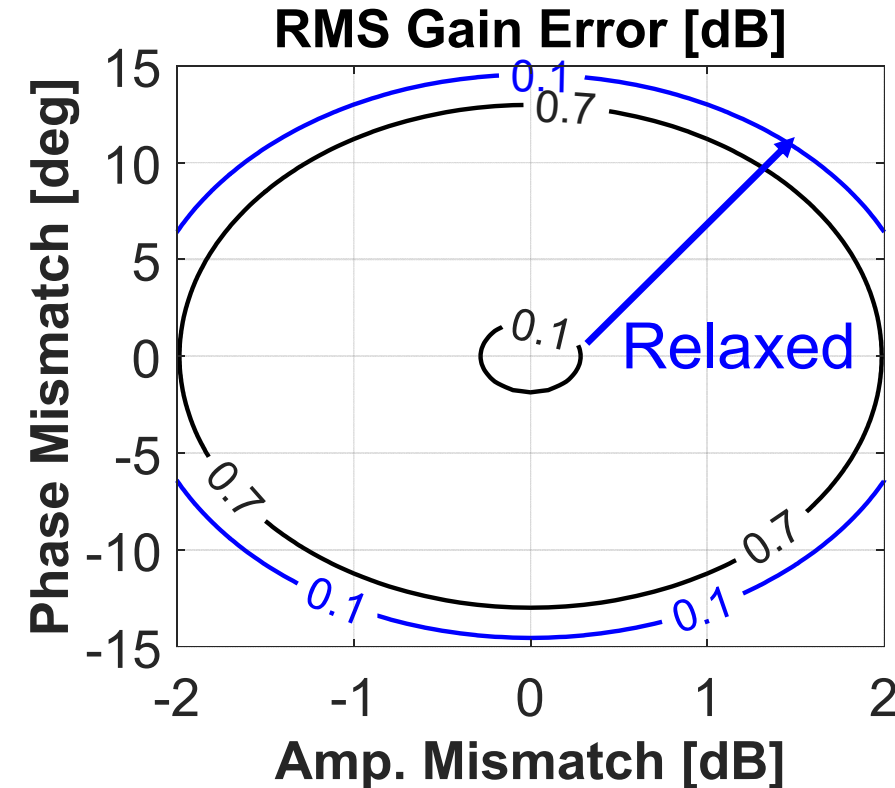
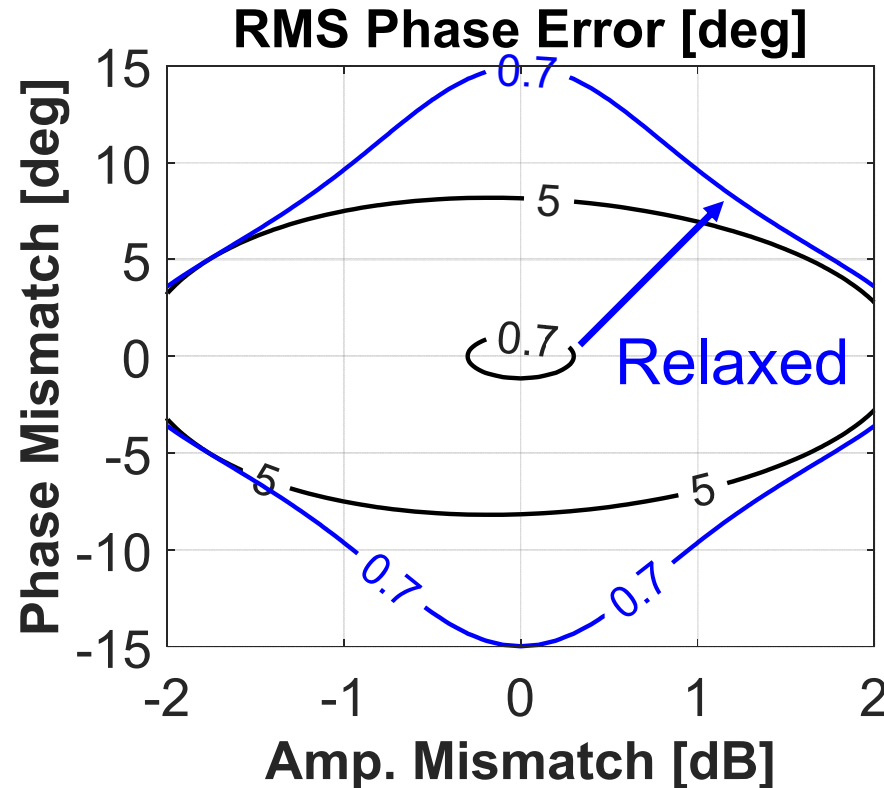
Nested VSPS Output:

w/i IQMM

$$\epsilon e^{j\theta} [0.5(\epsilon^{-1} e^{-j\theta} + \epsilon e^{j\theta}) G_I + jG_Q]$$



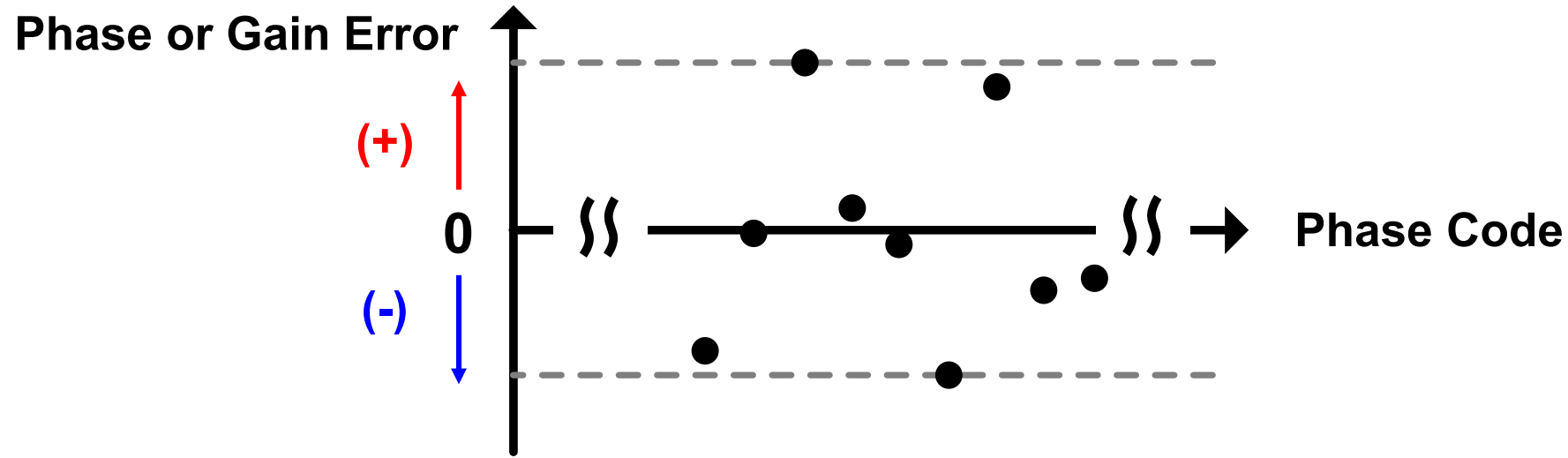
Nested Vector Modulation with IQMM



Proposed
Conventional

- Low RMS gain/phase error compared to conventional VSPS
- Nested VSPS is insensitive to IQMM

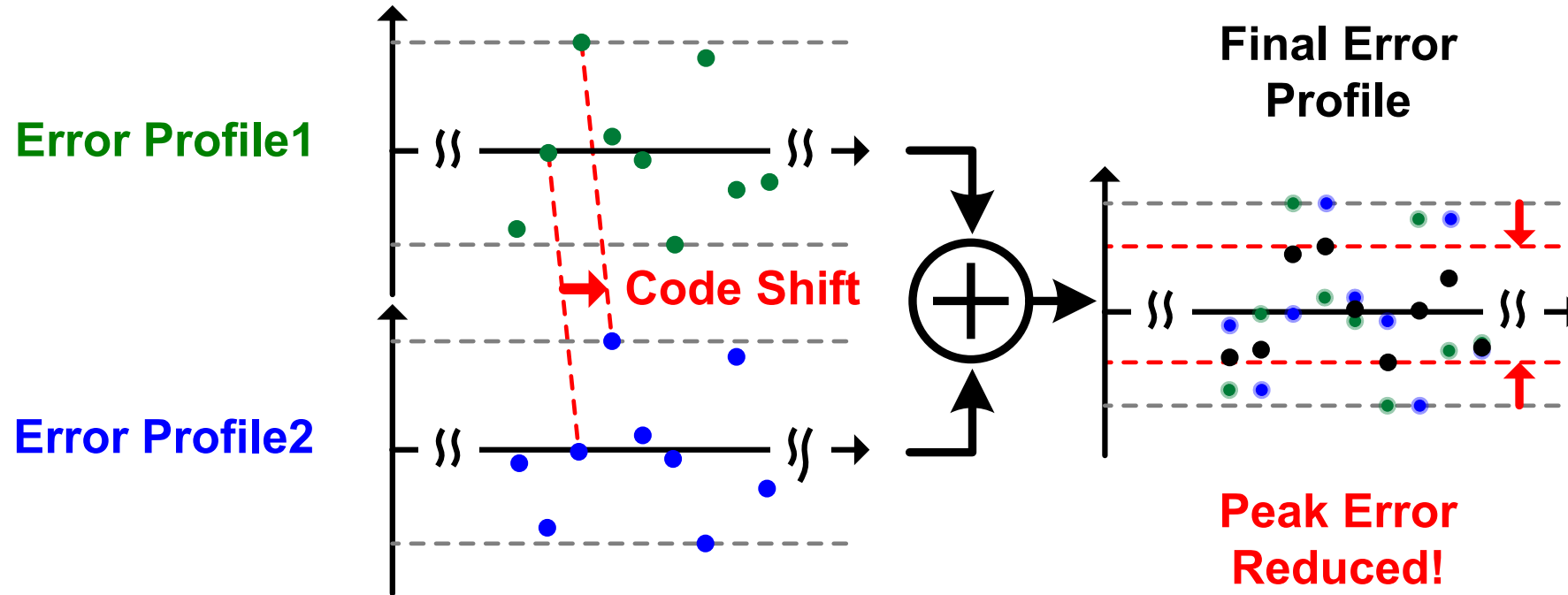
Peak Error Smoothing (PES)



- Usually, phase or gain error varies depending on the phase code of PS
- The main causes are IQMM, inaccurate IQ gain weighting and phase variation in VGA
- PVT variation further amplifies this phenomena

Peak Error Smoothing (PES)

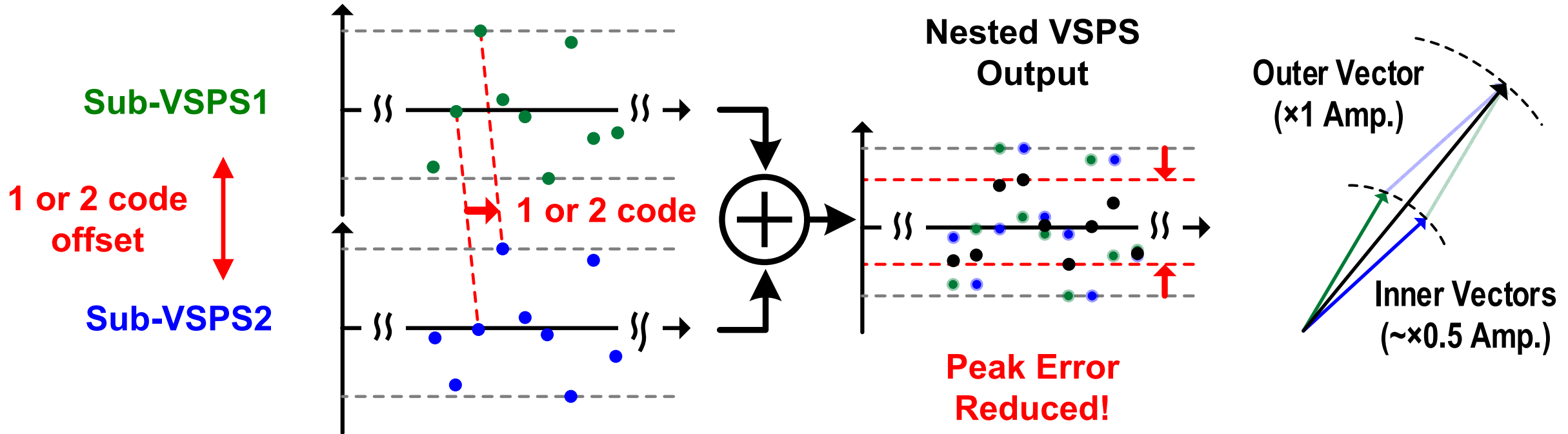
Concept:



- If the error plot is shifted and then averaged with the original error plot, the peak error can be reduced
- Phase-code-dependent errors can be reduced

Peak Error Smoothing (PES)

Realization:

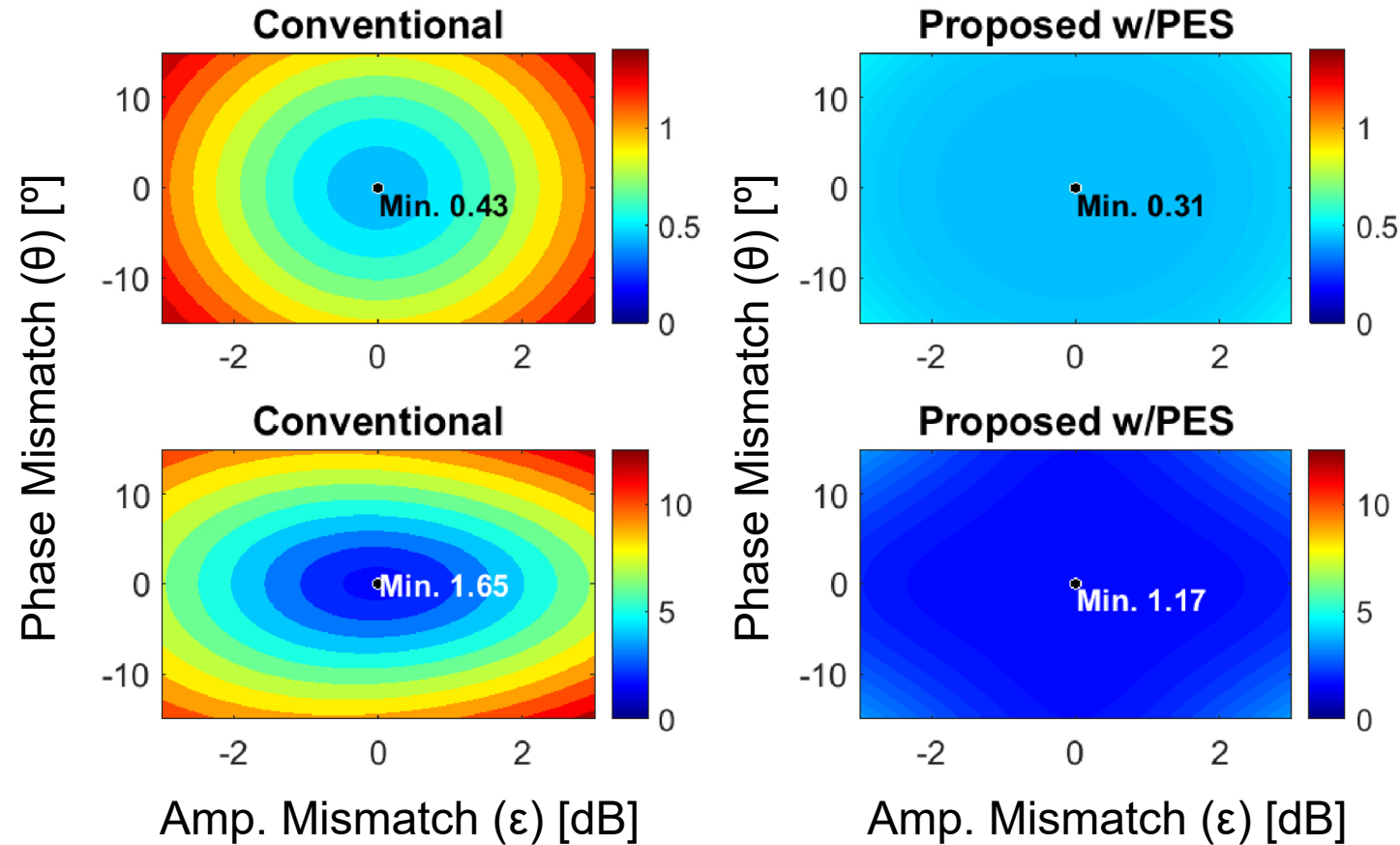


- Nested VSPS has two inner vectors with roughly $\frac{1}{2}$ magnitude
- Introducing a code offset between the two sub-VSPSs

Peak Error Smoothing (PES)

Under Max. $\pm 10\%$ G_I and G_Q Random Variation, 1000 Times Trial

Averaged RMS
Gain Error [dB]



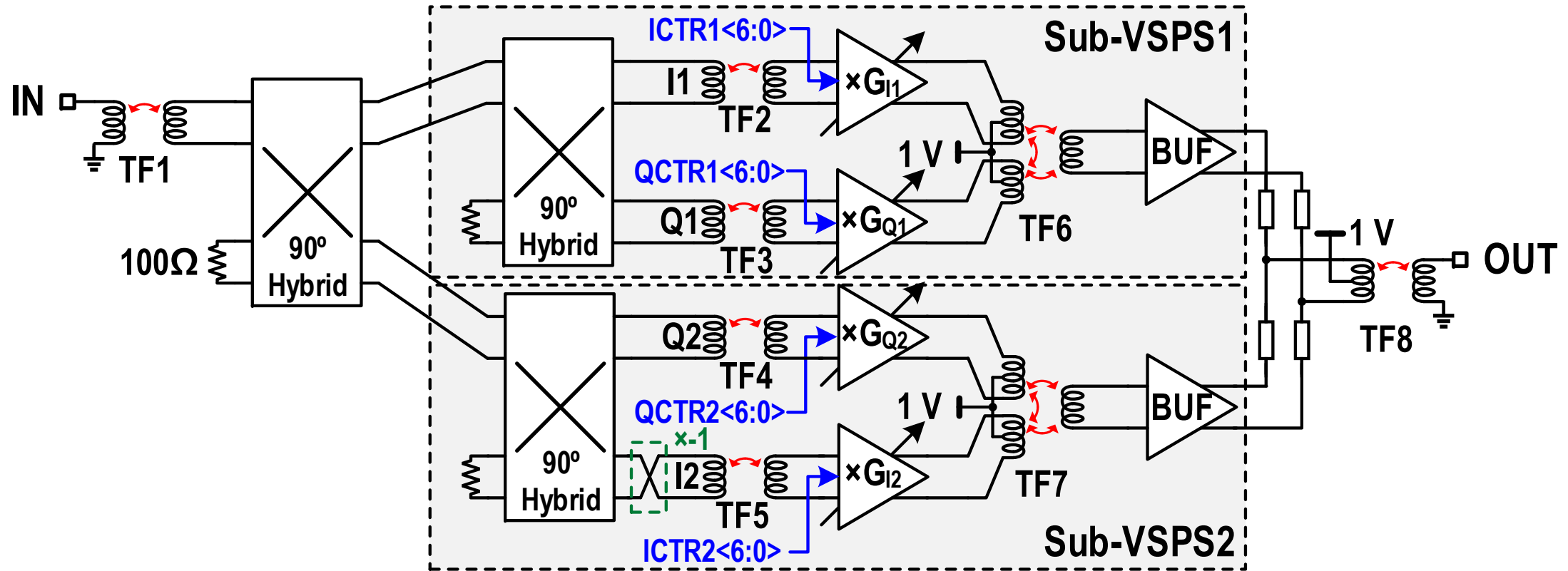
(Δ Phase code = 1)

■ **Error reduced, Still insensitive to IQMM**

Outline

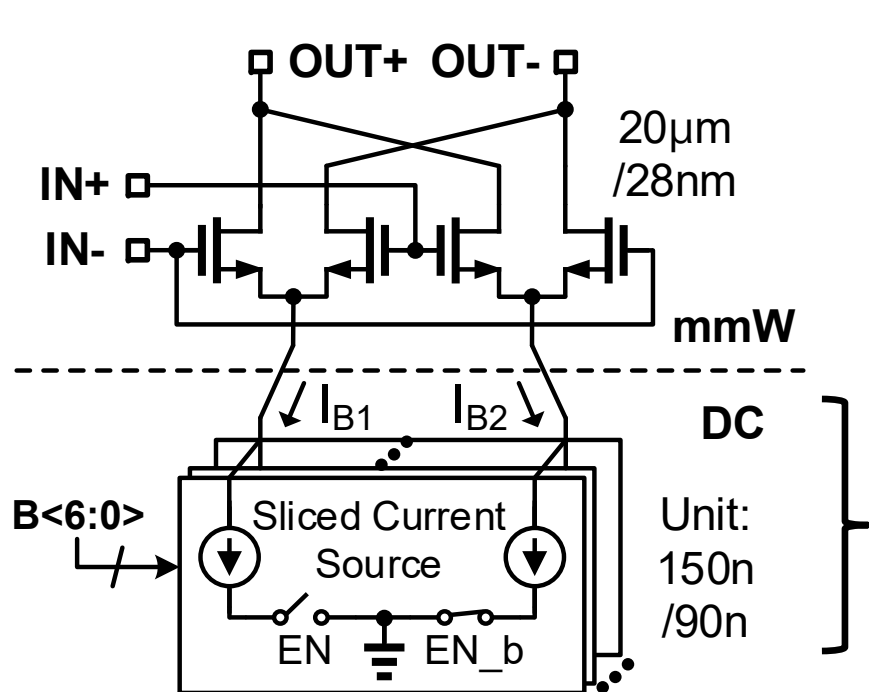
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Circuit Diagram



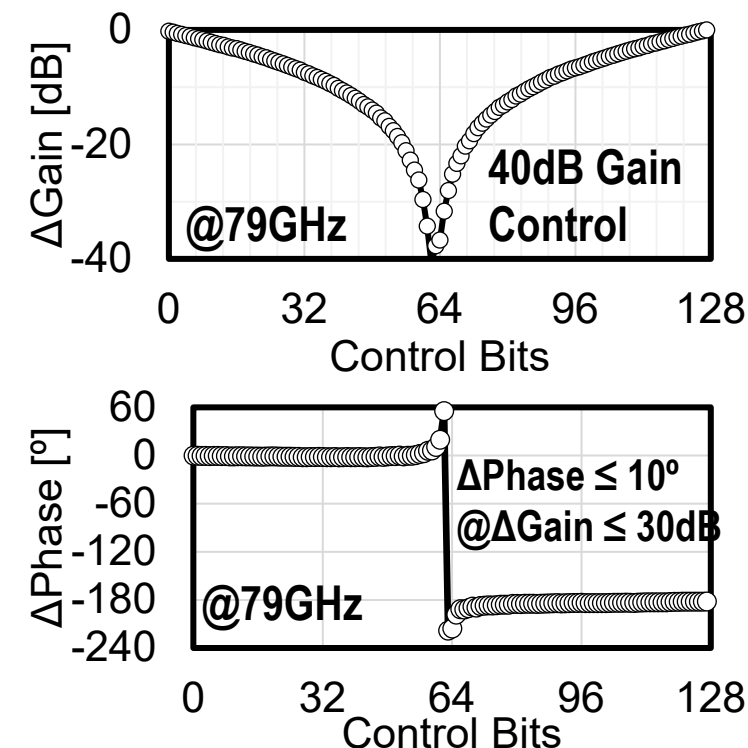
- IQ Generation using transformer-based 90° hybrid
- Two Sub-VSPSs with separated IQ control for PES

IQ Variable Gain Amplifiers



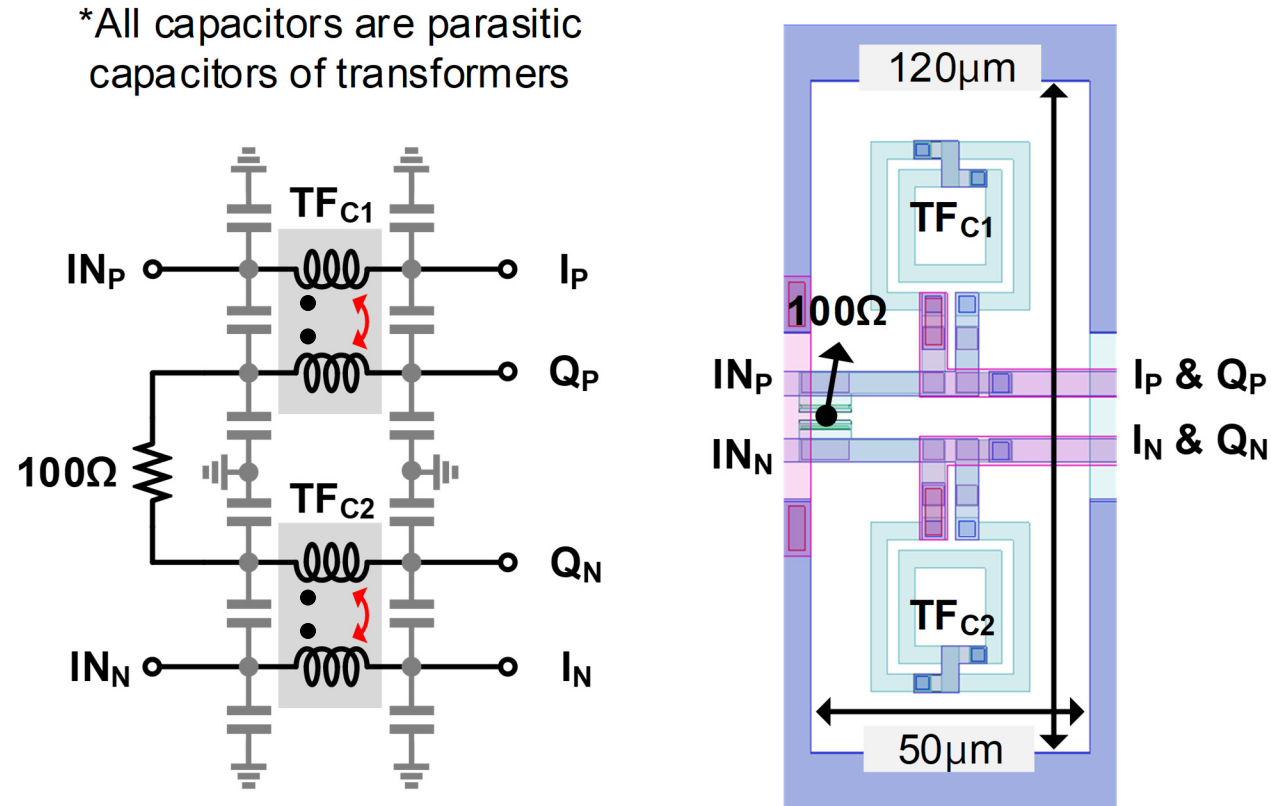
D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
D	D	D	D	D	D	A2	A1	A2	A1	D	D	D	D	D	D
D	D	A1	A2	B6	B6	B6	D	B6	B6	B6	B6	A2	A1	D	D
D	D	A2	A1	B6	B6	B6	B6	B6	B6	B6	B6	A1	A2	D	D
D	D	B6	B6	B5	B5	B5	B5	B5	B5	B5	B5	B6	B6	D	D
D	D	B6	B6	B5	B5	B4	B4	B4	B4	B5	B5	B6	B6	D	D
D	A1	B6	B6	B5	B4	B2	B3	B3	B2	B4	B5	B6	B6	A1	D
D	A2	B6	B6	B5	B4	B3	B0	B1	B3	B4	B5	B6	B6	A2	D
D	A1	B6	B6	B5	B4	B3	B1	D	B3	B4	B5	B6	B6	A1	D
D	A2	B6	B6	B5	B4	B2	B3	B3	B2	B4	B5	B6	B6	A2	D
D	D	B6	B6	B5	B5	B4	B4	B4	B4	B5	B5	B6	B6	D	D
D	D	B6	B6	B5	B5	B5	B5	B5	B5	B5	B5	B6	B6	D	D
D	D	A2	A1	B6	B6	B6	B6	B6	B6	B6	B6	A1	A2	D	D
D	D	A1	A2	B6	B6	B6	B6	B6	B6	B6	B6	A2	A1	D	D
D	D	D	D	D	D	D	A2	A1	A2	A1	D	D	D	D	D
D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D

A1: Always ON for I_{B1} , A2: Always ON for I_{B2}
D: Dummy



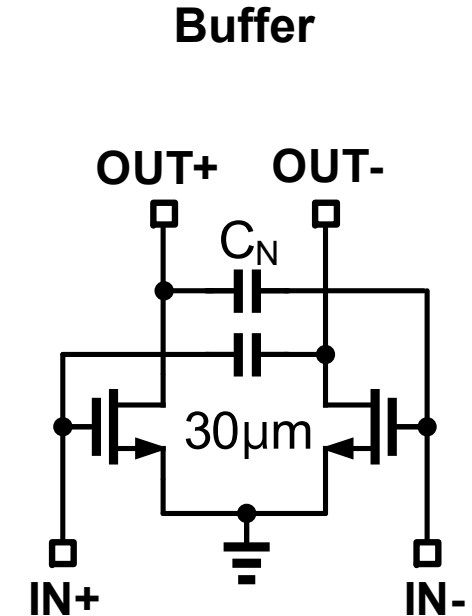
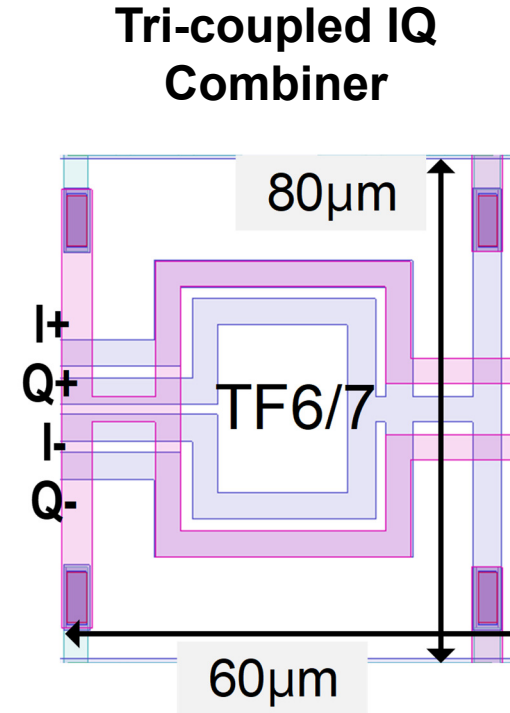
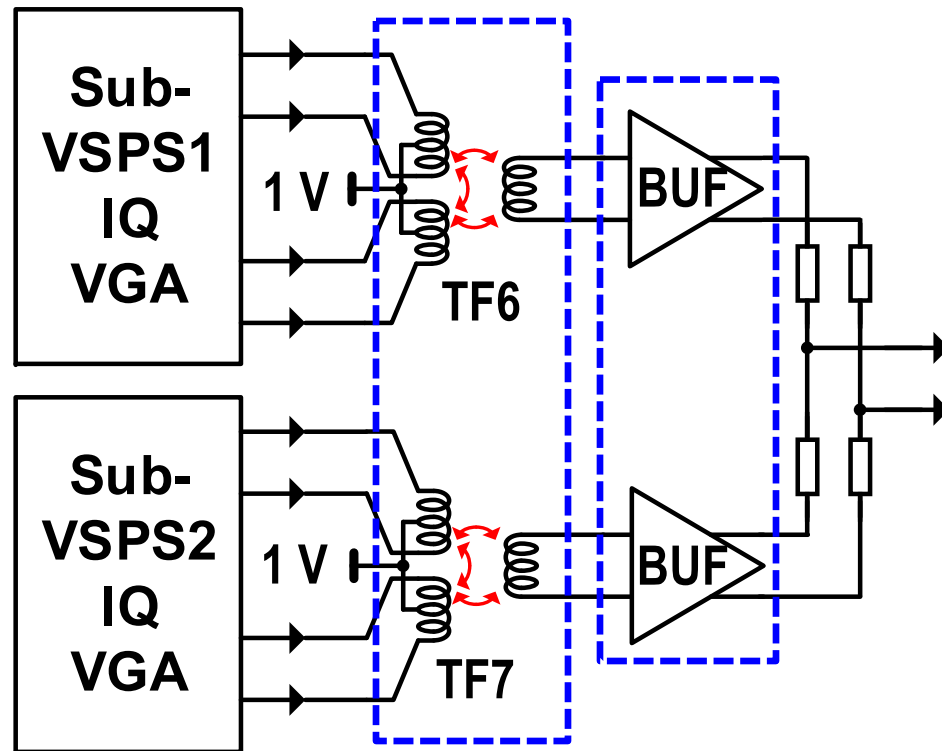
- Cross-Coupled CS with Sliced Current Source
- 40 dB Gain control range for high resolution phase shifting

IQ Generation



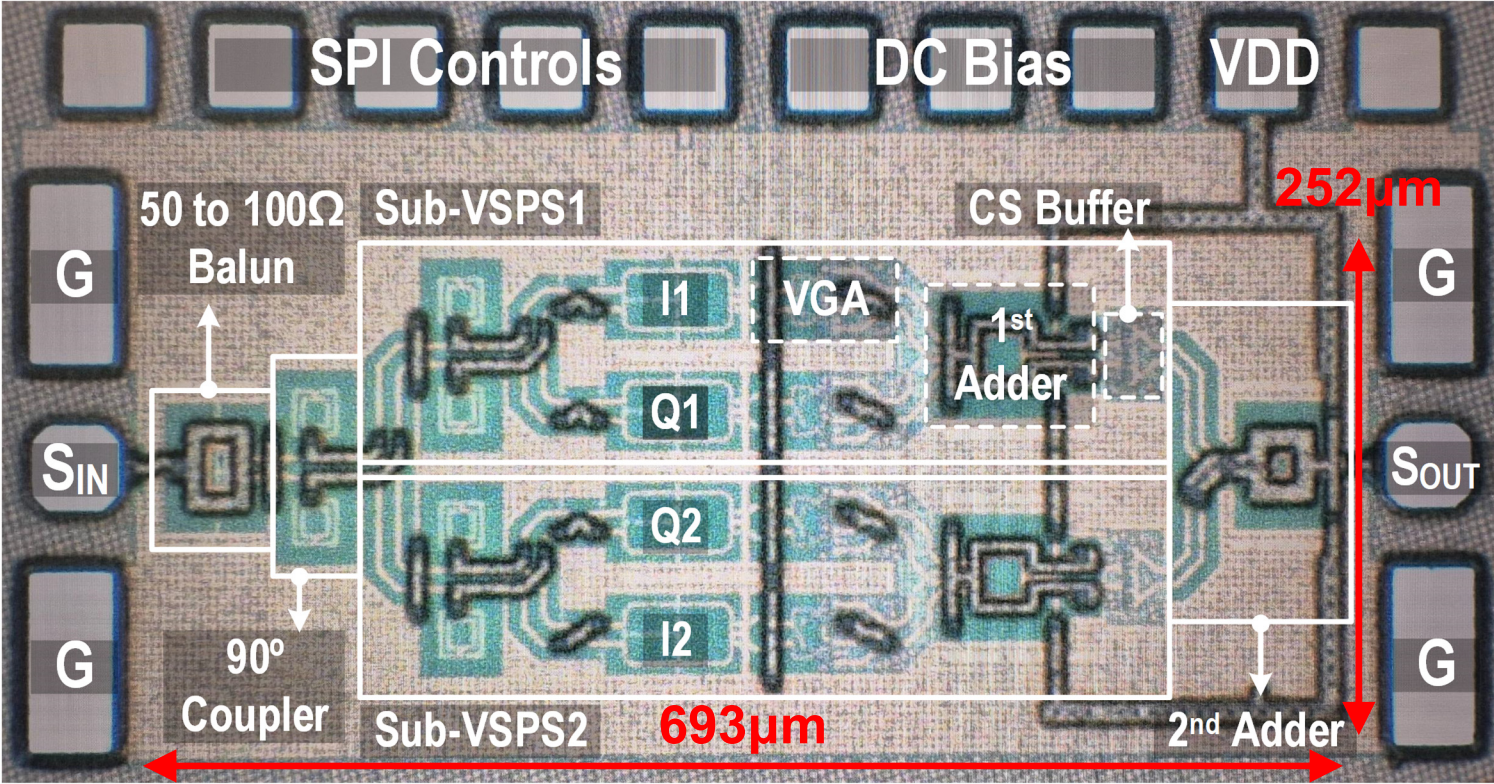
- Transformer-Based 90° Hybrid for compact implementation
- Amplitude mismatch ≤ 1 dB, Phase Mismatch $\leq 0.7^\circ$

Combiners and Buffers



- Tri-coupled transformer based combiner for compact size
- Neutralized CS buffer for high gain and isolation

Chip Micrograph and DC Breakdown



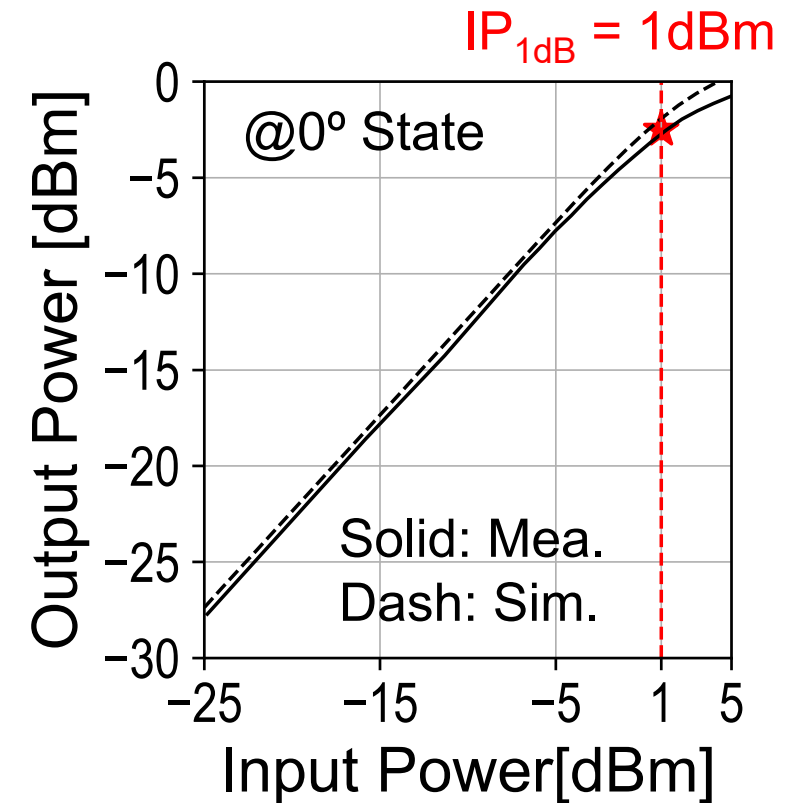
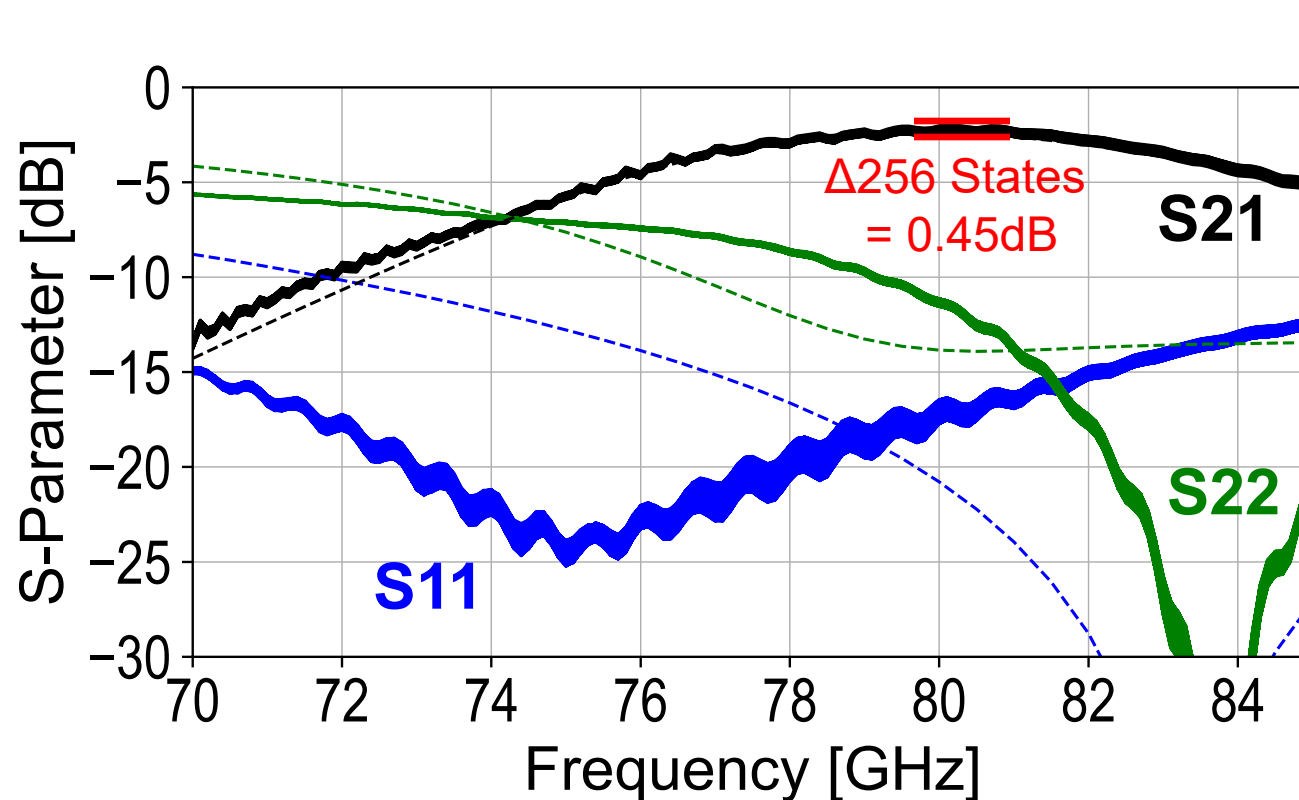
P_{DC}	Sub-VSPS1	I1-VGA	4mW	13.3%
		Q1-VGA	4mW	13.3%
		Buffer	7mW	23.4%
	Sub-VSPS2	Q2-VGA	4mW	13.3%
		I2-VGA	4mW	13.3%
		Buffer	7mW	23.4%
	Total		30mW	100%

- Fabricated in 28-nm CMOS Process
- Core Size: 693μm × 252μm

Outline

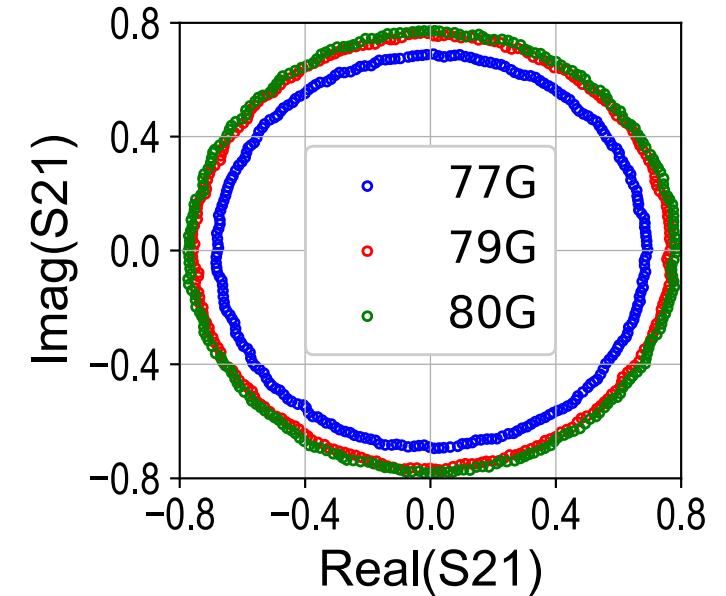
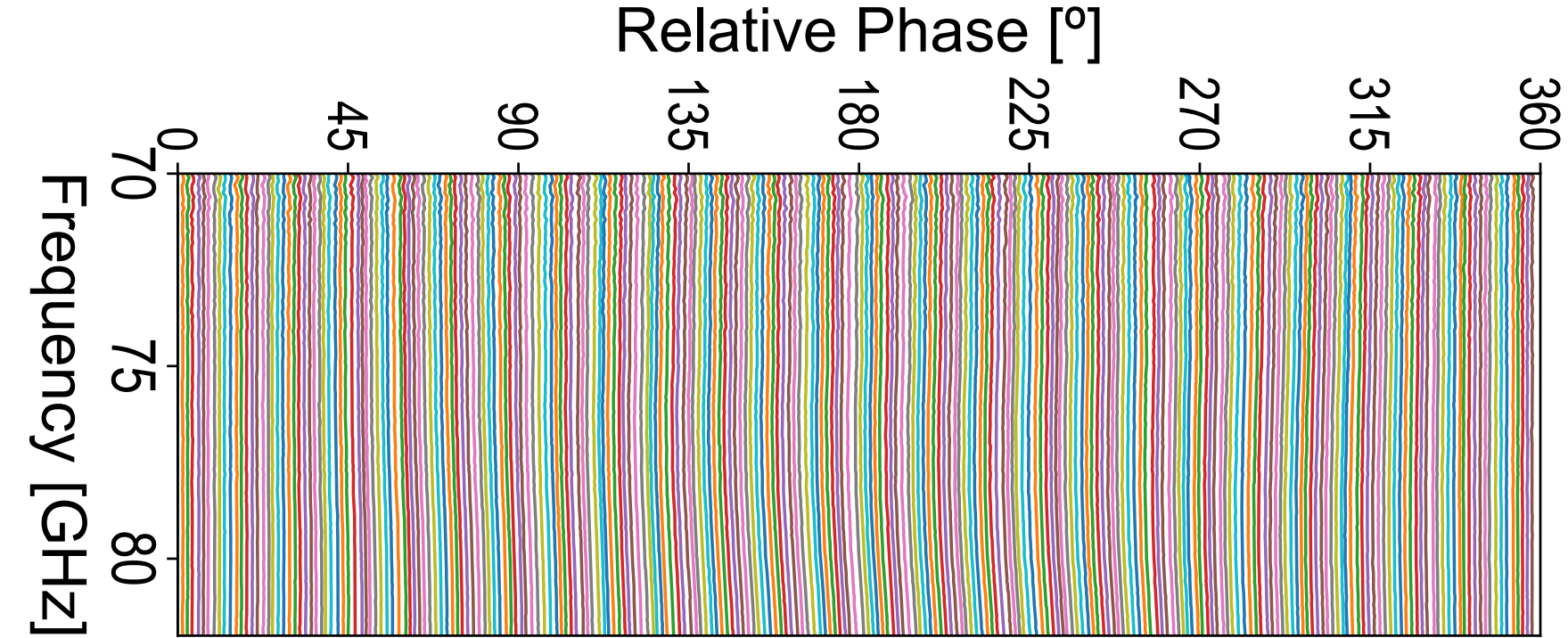
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S-Parameters and P1dB



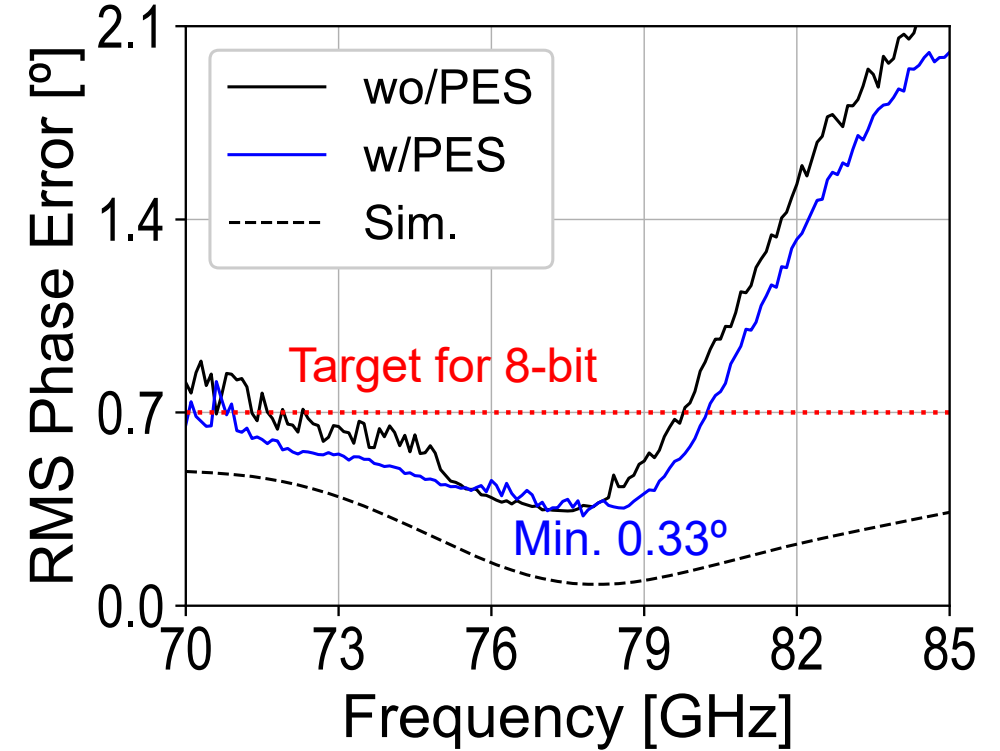
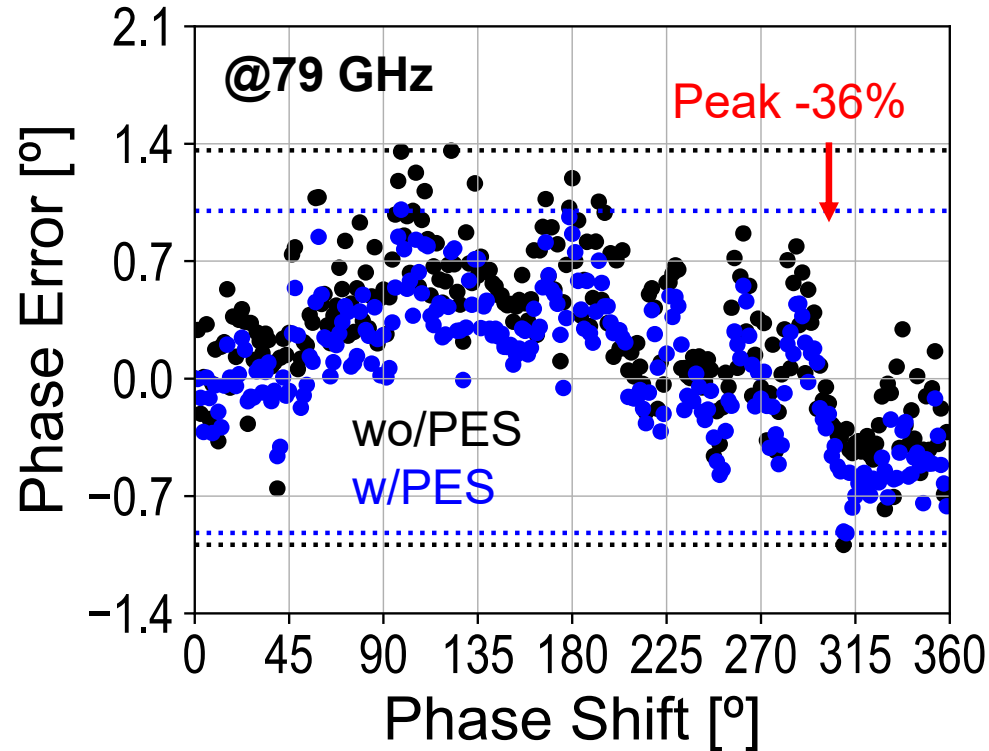
- Peak $G_{AVG} = -2.2 \text{ dB}$, $\Delta G \leq 0.45 \text{ dB}$ for 256 phase states
- $IP_{1dB} = 1 \text{ dBm}$ @ 0° Phase State

Relative Phase and Polar S21



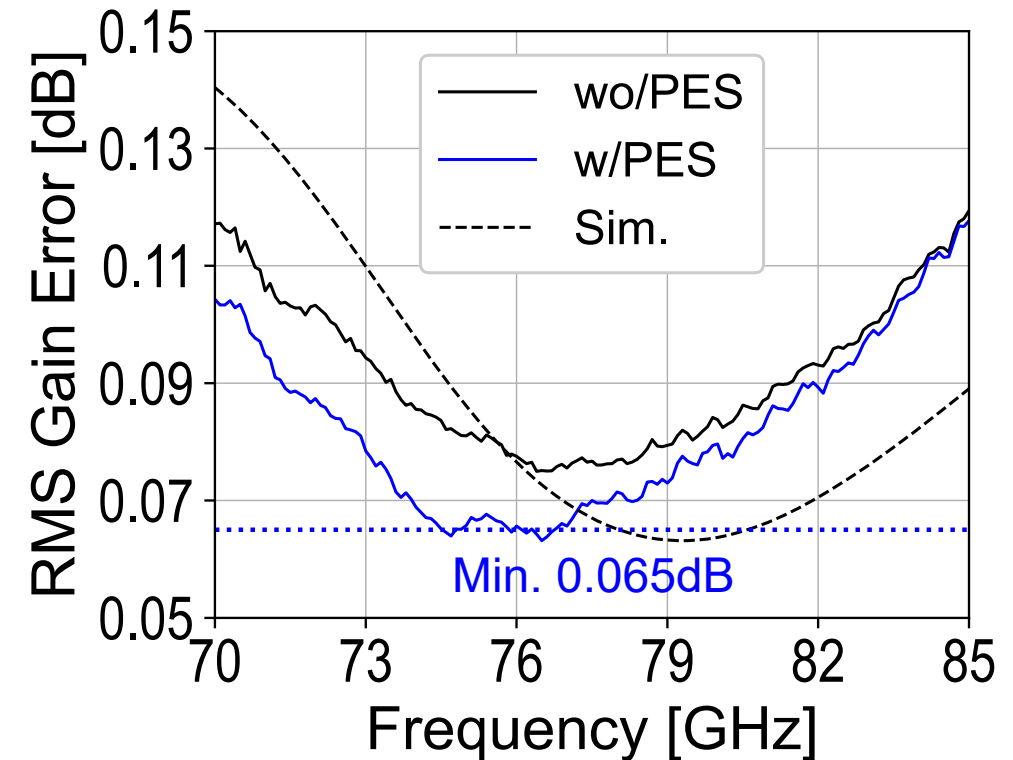
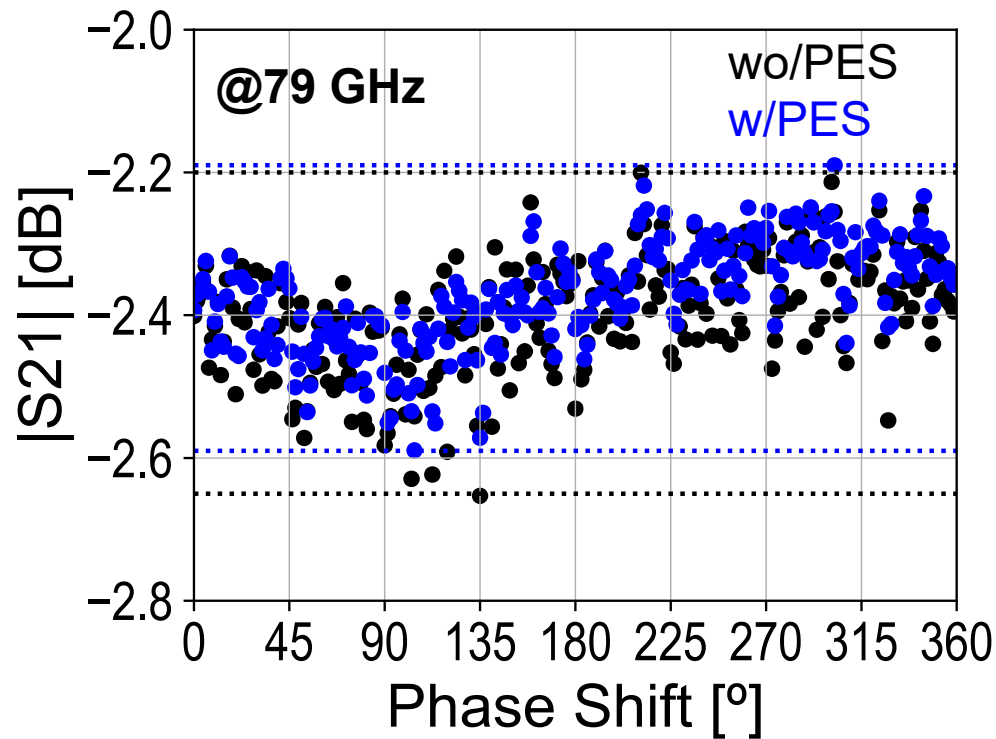
- Cover 360° phase shifting range with 8-bit resolution
- PES technique is adopted

Phase Error



- Maximum phase error is reduced owing to PES technique
- RMS phase error = 0.33° to 0.7° from 71 to 81 GHz

Gain Error



- Maximum gain error is also reduced owing to PES technique
- RMS gain error = 0.07 to 0.1 dB from 71 to 81 GHz

Comparison Table

	This Work	[2] JSSC 2017	[3] TMTT 2025	[6] ISSCC 2025	[4] IMS 2025	[5] TCAS-I 2021
Type	Active VSPS				Passive VSPS	
Architecture	Nested Structure	Conventional	Conventional	Hybrid Structure	Conventional	Conventional
Process	28nm CMOS	28nm CMOS FDSOI	0.13 μ m SiGe BiCMOS	55nm SiGe BiCMOS	28nm CMOS	40nm CMOS
Frequency [GHz]	71 to 81	78.7 to 92.8	91.5 to 98	125 to 170	64.7 to 93.3	70 to 90
Peak Average Gain [dB]	−2.2	2.3	14.4	−2.3	−11.8	−15.1
Phase Resolution [bit]	8 (1.41°)	4 (22.5°)	6 (5.62°)	5.3 (9°)	7 (2.81°)	6 (5.62°)
Gain Variation [dB]	< ± 0.23	$\pm 2.5^*$	$\pm 1^*$	$\pm 2^*$	$\pm 0.25^{**}$	< ± 0.6
RMS Gain Error [dB]	0.07 to 0.09	< 2	0.58 to 0.85	0.5 to 0.8	0.12 to 0.33	N/A
RMS Phase Error [°]	0.33 to 0.7	< 11.9	1.2 to 2.9	1 to 5	0.49 to 2	1.1 to 3.7
Error Reduction Technique?	YES (PES)	NO	NO	NO	NO	NO
IP1dB [dBm]	1	−7	−14	5.2 to 6.8***	N/A	N/A
DC Power [mW]	30	21.6	42	31	0	0
Core Size [mm ²]	0.17	0.12	0.33	0.2	0.15	0.15

*Graphically estimated **at 75GHz ***Calculated from OP1dB

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RMS Phase Error [°]	0.33 to 0.7	< 11.9	1.2 to 2.9	1 to 5	0.49 to 2	1.1 to 3.7
Error Reduction Technique?	YES (PES)	NO	NO	NO	NO	NO

→ 8-bit Resolution with 0.07 dB and 0.33° Min. RMS-Error
→ Best Phase Shifting Performance

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Conclusion

- **Nested VSPS architecture and PES technique were presented for high resolution PS design**
- **Nested VSPS performs vector modulation by splitting a single outer vector into two inner vectors**
- **Proposed architecture is insensitive to IQMM, and PES technique can reduce phase-code-dependent errors**
- **The implemented VSPS achieves 8-bit phase resolution with $0.33^\circ/0.07$ dB minimum RMS Errors**

Acknowledgment

- The authors would like to thank all members of the RF Development Team in Samsung Electronics for their valuable technical discussions and supports
- The authors also would like to thank all members in Samsung Semiconductor, US for their valuable technical discussions



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